COMSATS University Islamabad Vehari
Department of Computer Science
Assignment 2 – Spring 2026

Subject: Programming Fundamentals (CSC-103) **Instructor: Dr. Muhammad Mudassar**

Class BSCS Batch SP26 **Marks=10**

CLO-1 "Demonstrate the fundamental concepts of programming."

CLO-2 "Employ programming constructs using a programming language."

Instructions:

1. Attempt the pseudo code and flow chart questions on the paper or notebook and scan to add in the assignment
2. Create an integrated PDF of programs, outputs, and answers.
3. Use CamScanner or similar software for PDF creation.
4. It is mandatory to upload the PDF to the CUOnline portal.
5. Deadline: 27 March 2026 till 04:00 PM.

Solve the exercise questions. For your easiness I have placed the questions in the following

Question 1	The "Rock, Paper, Scissors" Logic You are designing a two-player Rock, Paper, Scissors game. • Input: Two characters ('R', 'P', 'S') representing Player 1 and Player 2's choices. • Requirement: Write a detailed pseudo code algorithm that determines the winner. Focus: Your solution must demonstrate the use of nested if-else or switch logic to handle all 9 possible combinations efficiently. Explain why you chose your specific structure over the alternative.
Question 2	Real-World System: Toll Booth Calculator A highway toll booth charges based on vehicle type. **Rates:** • Car: 70 Rupees • Bus: 150 Rupees • Truck: 400 Rupee **Input:** Vehicle type (String). **Task:** 1. Write pseudo code using a switch statement to handle the vehicle types. 2. Inside the switch, use Print to show the tax amount on the screen.
Question 3	Leap Year Validator A year is a leap year if it is divisible by 4. However, if it is divisible by 100, it is not a leap year, unless it is also divisible by 400.

	<u>Task:</u> Without writing actual Java code, write the pseudo code for a program that takes a year (e.g., 1900, 2000, 2024) and prints whether it is a leap year. **<u>Focus:</u>** Draw a flowchart (or describe the logic flow) that correctly implements this nested decision structure. Pay special attention to the order of your conditions.
Question 4	Exercise 4.18: (Credit Limit Calculator) Develop a Java application that determines whether any of several department-store customers has exceeded the credit limit on a charge account. For each customer, the following facts are available: a) account number b) balance at the beginning of the month c) total of all items charged by the customer this month d) total of all credits applied to the customer's account this month e) allowed credit limit. The program should input all these facts as integers, calculate the new balance (= beginning balance + charges – credits), display the new balance and determine whether the new balance exceeds the customer's credit limit. For those customers whose credit limit is exceeded, the program should display the message "Credit limit exceeded".
Question 5	4.19 ***(Sales Commission Calculator)*** A large company pays its salespeople on a commission basis. The salespeople receive $200 per week plus 9% of their gross sales for that week. For example, a salesperson who sells $5,000 worth of merchandise in a week receives $200 plus 9% of $5000, or a total of $650. You've been supplied with a list of the items sold by each salesperson. The values of these items are as follows: Item Value 1 239.99 2 129.75 3 99.95 4 350.89 Develop a Java application that inputs one salesperson's items sold for last week and calculates and displays that salesperson's earnings. There's no limit to the number of items that can be sold.
Question 6	4.20 ***(Salary Calculator)*** Develop a Java application that determines the gross pay for each of three employees. The company pays straight time for the first 40 hours worked by each employee and time and a half for all hours worked in excess of 40. You're given a list of the employees, their number of hours worked last week and their hourly rates. Your program should input this information for each employee, then determine and display the employee's gross pay. Use class Scanner to input the data.
Question 7	4.21 ***(Find the Largest Number)*** The process of finding the largest value is used frequently in computer applications. For example, a program that determines the winner of a sales contest would input the number of units sold by each salesperson. The salesperson who sells the most units wins the contest. Write a pseudocode program, then a Java application that inputs a series of 10 integers and determines and prints the largest integer. Your program should use at least the following three variables: a) counter: A counter to count to 10 (i.e., to keep track of how many numbers have been

	input and to determine when all 10 numbers have been processed). b) number: The integer most recently input by the user. c) largest: The largest number found so far.

Question 8	4.28 ***(Another Dangling-***else ***Problem)*** Modify the given code to produce the output shown in each part of the problem. Use proper indentation techniques. Make no changes other than inserting braces and changing the indentation of the code. The compiler ignores indentation in a Java program. We've eliminated the indentation from the given code to make the problem more challenging. [*Note:* It's possible that no modification is necessary for some of the parts.]
	`if (y == 8)`
	`if (x == 5)`
	`System.out.println("@@@@@");`
	`else`
	`System.out.println("#####");`
	`System.out.println("$$$$$");`
	`System.out.println("&&&&&");`
	a) Assuming that x = 5 and y = 8, the following output is produced:
	@@@@@
	$$$$$
	&&&&&
	b) Assuming that x = 5 and y = 8, the following output is produced:
	@@@@@
	c) Assuming that x = 5 and y = 8, the following output is produced:
	@@@@@
	d) Assuming that x = 5 and y = 7, the following output is produced. [*Note:* The last three output statements after the else are all part of a block.]
	#####
	$$$$$
	&&&&&

Question 9	5.12 (Calculating the Product of Odd Integers) Write an application that calculates the product of the odd integers from 1 to 15
Question 10	Write a program that asks the user to enter a number and then using a loop print table for that number